NAAN MUDHALVAN PROJECT DOCUMENTATION

Student Enrollment & Course Management System (SECMS)

Submitted by,

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Year : 3rd yr

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Submitted To,

Naan Mudhalvan Pogram

Acadamic Year : 2025-2026

User Story:

The Student Enrollment and Course Management System is a Salesforcedrivenapplication designed to enhance and automate the management of academic enrollmentprocesses in educational institutions. The system provides a centralized platform whereadministrators, enrollment officers, and instructors can efficiently maintain student data,manage course offerings, process enrollments, assign instructors,track payments, andgenerate reports. By

integrating these operations within a single Salesforce environment,the system improves efficiency, data accuracy, and overall academic administration.

1. INTRODUCTION

1.1 Project Overview :

The Student Enrollment & Course Management System is a cloud-based application management. developed using Salesforce to manage student academic activities efficiently
.The system helps educational institutions organize and maintain information about students,courses,instructors, and enrollments in a structured manner.Through this system, administrators and enrollment officers can easily register students,manage course details, assign instructors and monitor enrollment activities
By using Salesforce reports and dashboards, management can track important data such as student enrollments, course demand, instructor assignments, and fee payments. This system reduces manual work and improves the accuracy of academic operations

1.2 Purpose :

The pupose of this project is to build a digital asystem that simplifies the student enrollment process.Many institutions still manage enrollment data manually using speadsheets.

Automated enrolment proccesing
Secure access control for different users
Real time reports and dashboards

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Enrollment Officer	manage student enrollments records	data is stored in different places	Manual systems are used	Stressed and time consuming
PS-2	Administrator	Monitor course registration and revenue	Reports are not generated	data is not centralized	Difficult to make decisions

2.2 Empathy Map Canvas

User Persona: Enrollment

Officer

- **Think & Feel:** Wants to manage student enrollments without mistakes
- **Hear:** Complaints from students about delays. • **See:** Multiple spreadsheets and manual records.
- **Say & Do:** checks records repeatedly to avoid errors.
- **Pain:** Duplicate entries and slow approval process
- **Gain:** Automated,system with dashboards quick approvals

2.3 Brainstorming & Idea Prioritization

Step 1 – Brainstormed Ideas

- Spreadsheet-based tracking system
- Web-based enrollment portal
- ERP system integration
- Salesforce-based management system

Step 2 – Grouping

- Semi-automated solutions
- Manual solutions
- Fully automated cloud solutions

Step 3 – Prioritization

Salesforce-based SECMS was selected due to **high feasibility** and **high impact**.

3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map Stages

covered:

- Student registration
- Course Exploration
- Enrollment Request
- Payment Verification
- Approval Process
- Instructor Allocation
- Monitoring & Reporting

3.2 Solution Requirements

Functional Requirements

FR No	Epic	Description
FR-1	Student Management	Maintain student profiles
FR-2	Course Management	Create and update courses information
FR-3	Enrollment Mangement	Process course enrolment requests
FR-4	Instructor Allocation	Assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR-1: Usability

The application should provide a user-friendly interface that allows users to easily navigate between different modules and perform tasks with minimal training.

Examples:

- Role-based dashboards
- Quick student registration form
- One-click enrollment approval

NFR-2: Security

The system must protect sensitive student information by implementing authentication and role-based access control.

Examples:

- Instructors view only assigned students
- Only admins can modify course details
- Students cannot access other student records

NFR-3: Reliability

The system should perform consistently without errors during enrollment processing and data updates.

Examples:

- No duplicate course registration
- Accurate status updates after approvals • Consistent report generation

NFR-4: Performance

The application should provide quick response times even when handling a large number of student records.

Examples:

- Fast search for student records
- Quick dashboard refresh
- No lag during bulk uploads

NFR-5: Scalability

The system must supp number of students, c 'uture growth in the ↓ s,

and departments **Examples:**

- Extend to mobile or portal integration
- Add new departments without redesign
- Handle increased admissions annually

NFR-6: Availability

Users should be also to access the system online with minimal

downtime during operational hours

Examples:

- Reliable during peak admission season
- Automatic backup of student records
- 24/7 cloud access

3.3 Data Flow Diagram

- User inputs student and enrollment data
- System validates and stores records
- Approval process routes data
- Project planning tables
- Solution architecture
- Reports and dashboards consume processed data.

The system processes enrollment requests, routes approvals, validates data, assigns instructors, and generates reports.

3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning experience
Logic	Salesforce Flows builder
Database	Salesforce custom Objects
Automation	Validation Rules and Approval Process
Reporting	Reports and Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Many educational institutions still rely on manual methods such as spreedsheets,emails and paper records to manage student enrolment and course information.

The application allows administrators to maintain student records, create and manage course information, assign instructors, and monitor enrollment requests. Automation tools available in Salesforce help reduce manual effort by validating data, triggering workflows, and handling approval processes.

The system also supports reporting and dashboard features, enabling management to analyze important metrics such as course registrations, student activity, and fee collections

4.2 Proposed Solution

Functional Requirements

Parameter	**Description**
Problem	Manual handling of enrollment data
Solution	Cloud based system built on Salesforce
Novelty	Automated workflows
Impact	Improved student experience

Revenue Model	Institutional deployment
Scalability	Multi-department support

5. Project Development Phase:

Project Flow:

Milestone 1: Salesforce Developer Account Creation
Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)
Milestone 3: Tabs Creation
Milestone 4: Lightning App Creation
Milestone 5: Fields Creation
Milestone 6: Create Relationships
Milestone 7: Page Layout Customization
Milestone 8: Validation Rules
Milestone 9: Flows (Automation)
Milestone 10: Approval Process
Milestone 11: Reports
Milestone 12: Dashboards
Milestone 13: Security Setup
Milestone 14: Conclusion

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows

6. Creating approval workflows
7. Developing reports & dashboards
8. Setting up security with profiles, permission sets, and sharing
9. End-to-end application testing

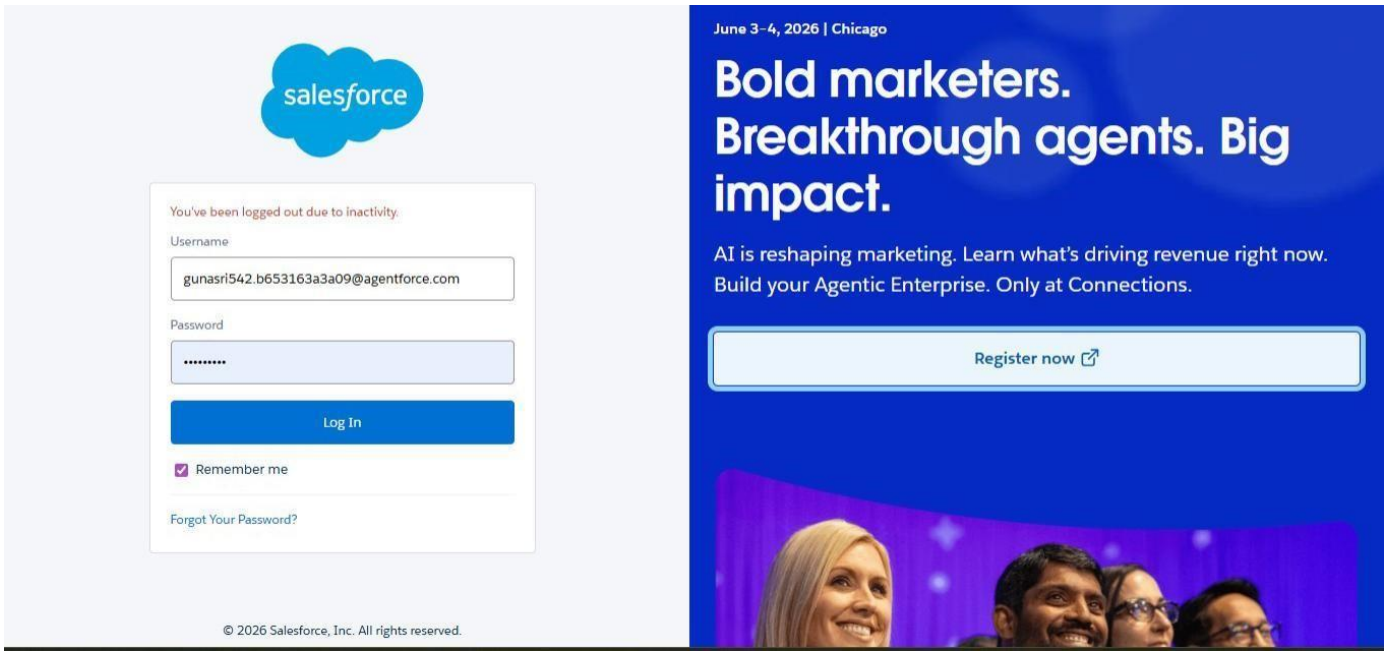
Milestone 1-Salesforce :

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

Go to https://developer.salesforce.com/signup

On the sign up form, enter the following details :



First name & Last name

Email

Role : Developer

Company : College Name

County : India

Postal Code : pin code

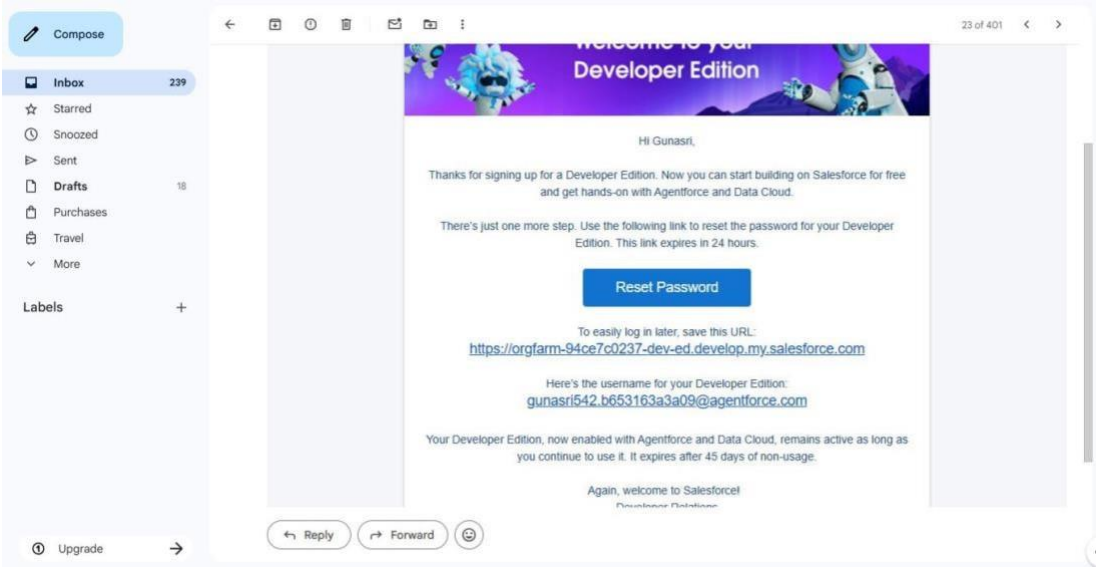
Username : should be a combination of your name and company

This need not be an actual email id, you can give anything in the format :

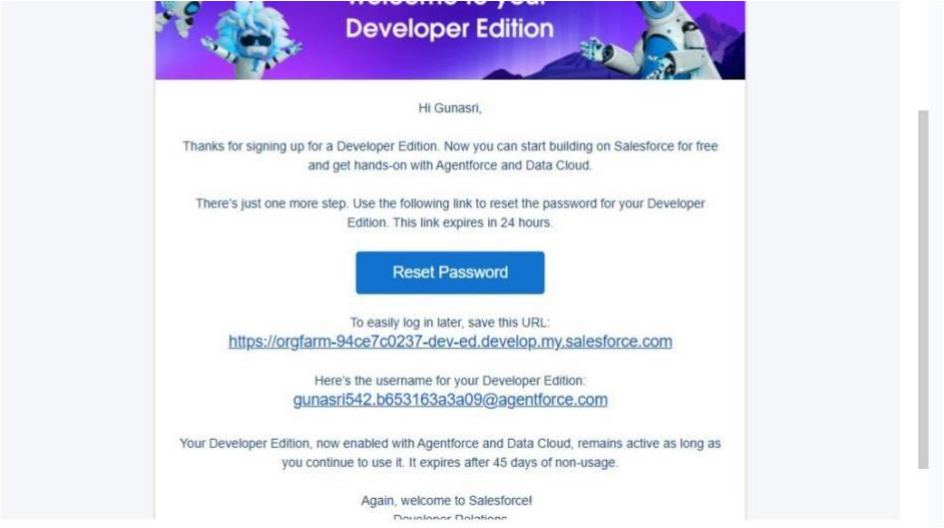
username@organization.com I
used gunasri542@gmail.com
Click on sign me up after filling these.

Activity 2: Account Activation:

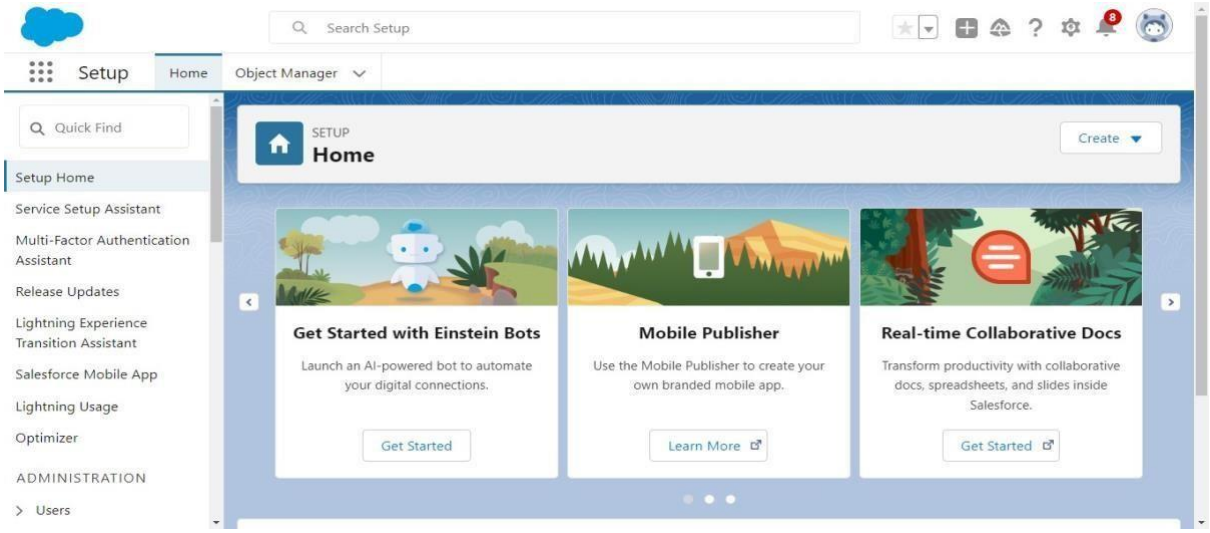
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.



2. Click on Verify Account
3. Give a password and answer a security question and click on change password.



4. Then you will redirect to your salesforce setup page.



Milestone 2 – Objects:

Activity 1: Creating Student Object:

The purpose of creating the Student custom object is to store and manage information about students such as their personal details, contact information, and enrollment status.

To create the Student object:

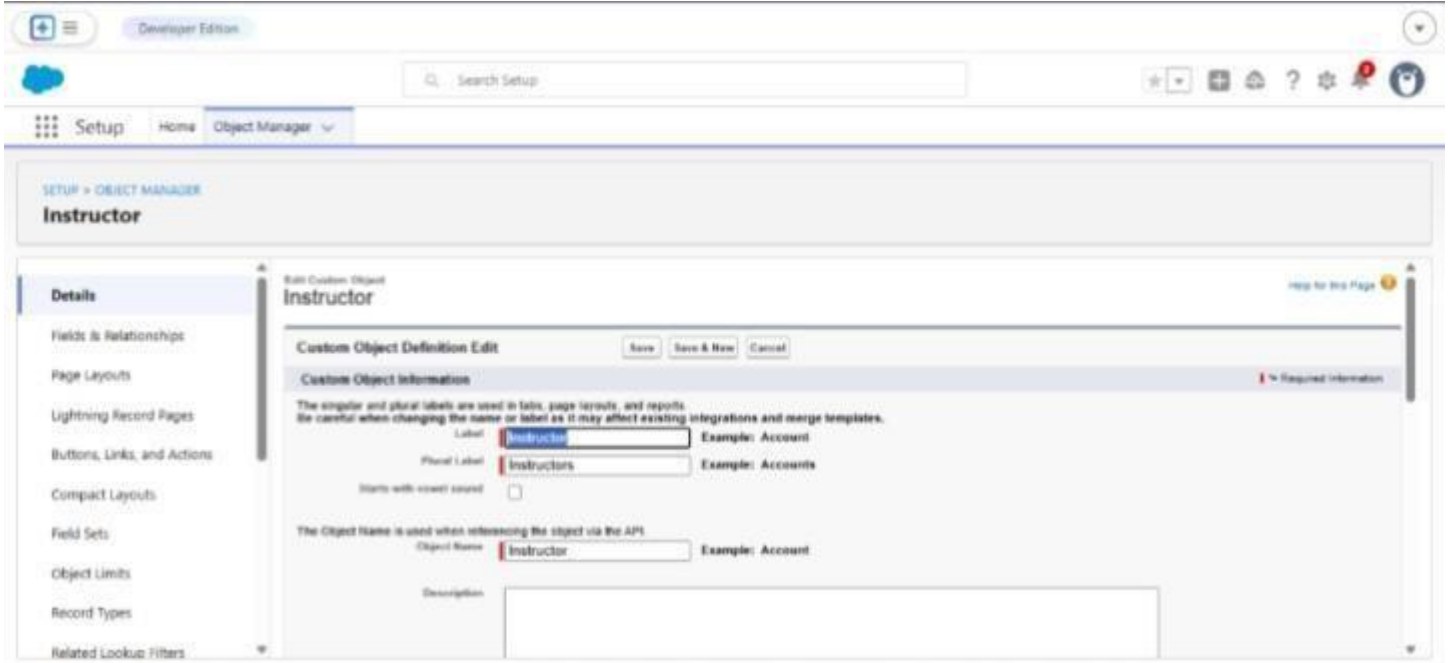
Go to the Setup page.

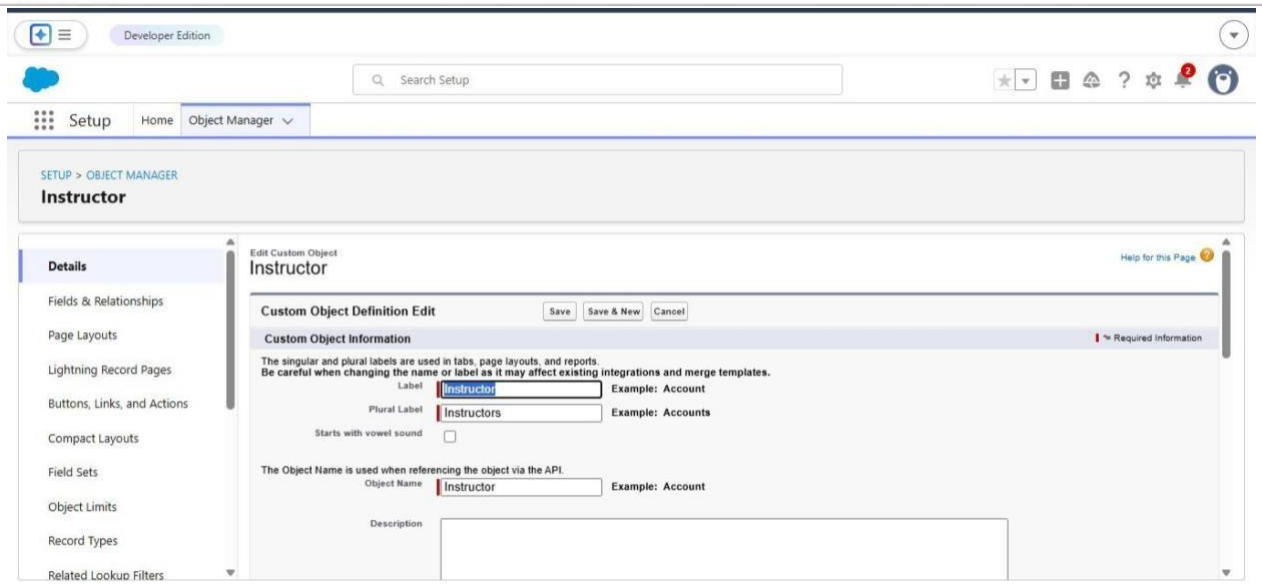
Click on Object Manager.

Click on Create → Custom Object.

Enter the details as below:

1. Label: Student
2. Plural Label: Students
3. Object Name: Student
4. Record Name: Student Name
5. Data Type: Text
6. Under Optional Features, select:
7. Allow Reports
8. Allow Activities (optional but recommended) 9. Track Field History (optional, if you want)
10. Under Search Status, select: 11. Allow Search
12. Click Save.





In the same way Create **Courses, Instructors,** and **Enrolments objects**

Milestone 3 - Fields:

Table 1: Student Object Fields:

Field Label	Data Type
Student Name	Text (Standard)
Email	Email
Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Field Label	Data Type
Course Name	Text (Standard)
Category	Picklist
Duration	Number/Text

Course Fee	Currency
Description	Text Area

Table 2: Course Object Fields:

Table 3: Instructor Object Fields:

Field Label	**Data Type**
Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone

Email	Email

Table 4: Enrollment Object Fields:

Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date

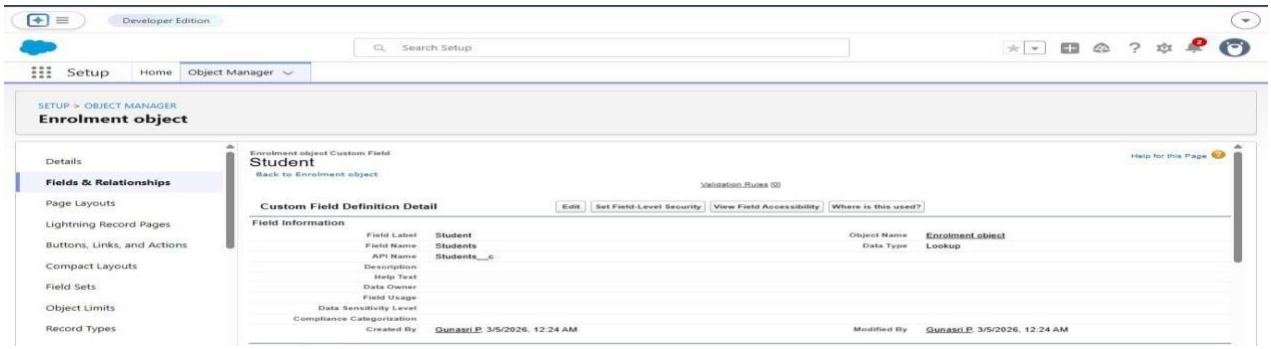
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Milestone 4 – Create Relationships:

1. **Activity 1: Create Lookup from Enrollment to Student:**

Go to Setup → click Object Manager.

2. Type Enrollment in Quick Find → click Enrollment__c object.
3. Click Fields & Relationships → click New.
4. Select Lookup Relationship → click Next.
5. For Related To, select Student__c → click Next.
6. Enter Field Label: Student (API Name: Student__c).
7. Click Next through Field-Level Security and Layout options (ensure it's visible and added to layout).
8. Click Save.

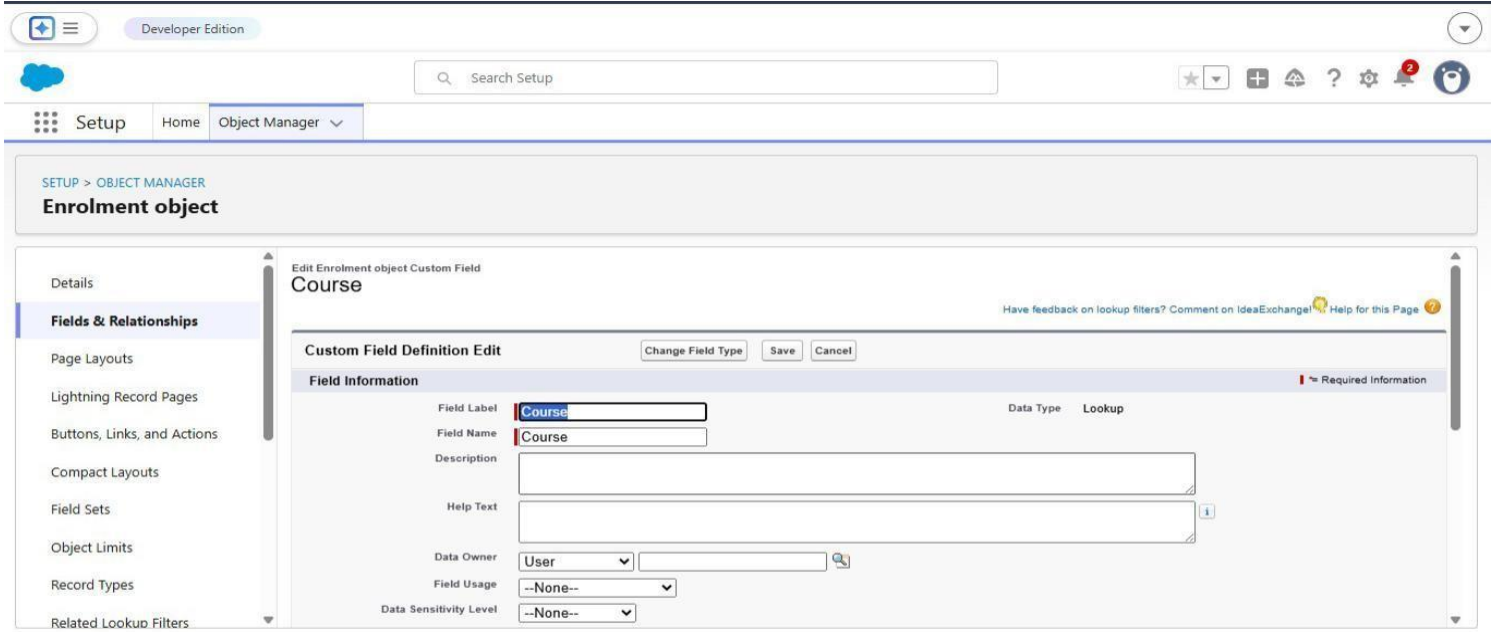


Activity 2: Create Lookup from Enrollment to Course

Purpose:
To link each Enrollment to a Course.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Select Lookup Relationship → Next.
4. Related To: select Course__c → click Next.
5. Field Label: Course (API Name: Course__c).
6. Click Next (set visibility and layout) → Save.



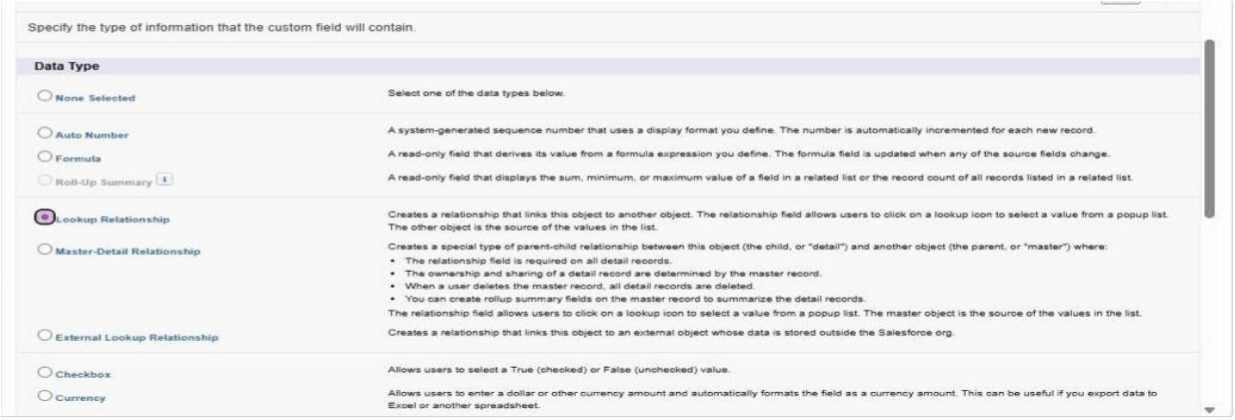
Activity 3: Create Lookup from Enrollment to Instructor:

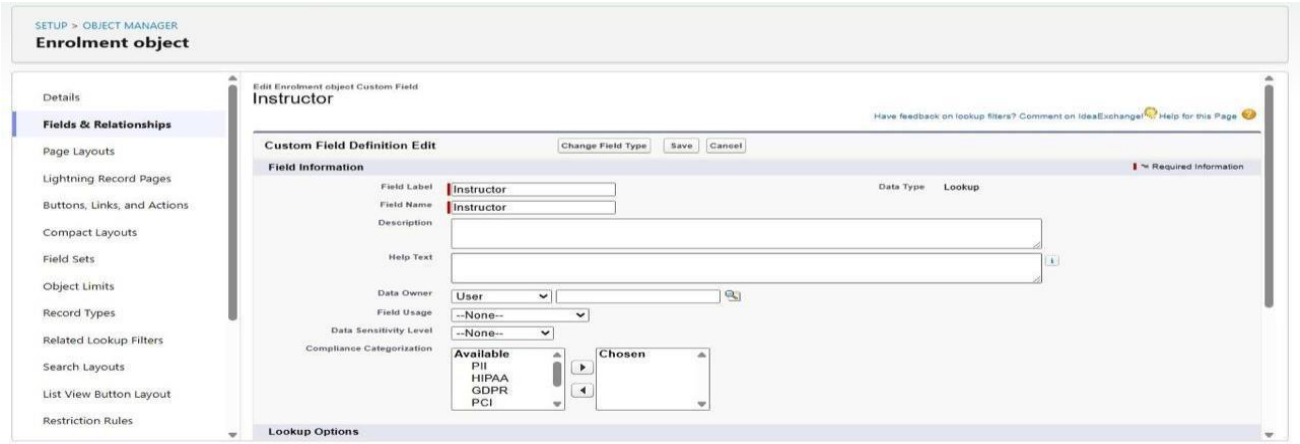
Purpose:

To store which Instructor is assigned to a particular Enrollment.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Choose Lookup Relationship → Next.
4. Related To: select Instructor__c → Next.
5. Field Label: Instructor (API Name: Instructor__c).
6. Complete Field-Level Security and Page Layout options → Save





Milestone 5 – Validation Rules:

Activity 1: Student Email Domain Validation

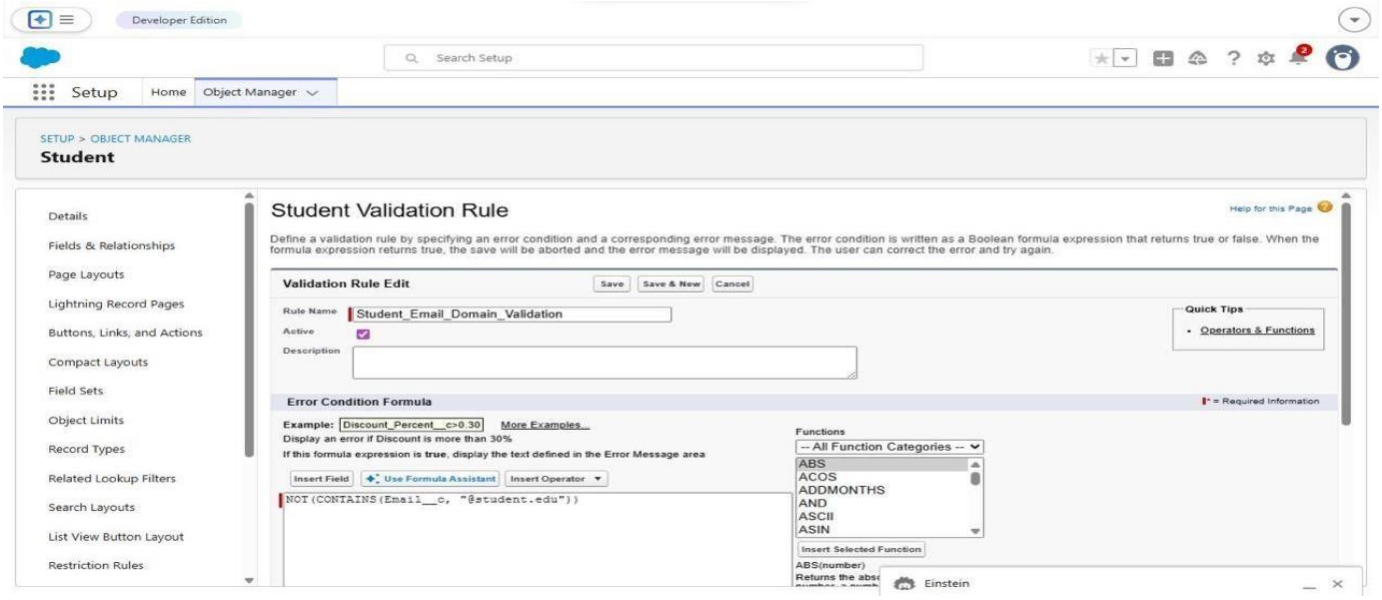
Purpose:
To ensure students use only allowed institutional email addresses.

Steps:

Go to Setup → Object Manager → Student__c.
Click Validation Rules → New.
Rule Name: Student_Email_Domain_Validation.
Error Condition Formula: (example for @student.edu)

- NOT(CONTAINS(Email__c, "@student.edu"))

Error Message: Email must be a valid student institutional email.
Error Location: Field → Email__c. Click Save.



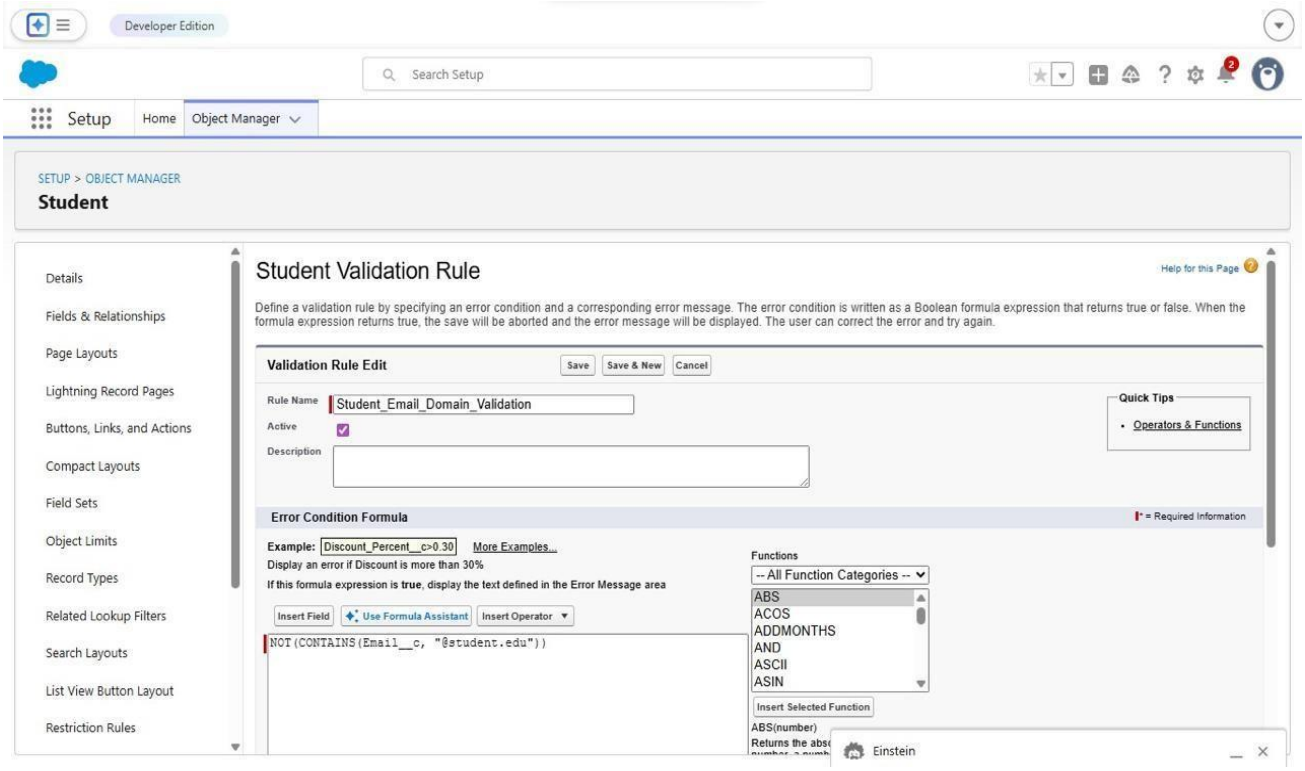
Activity 2: Fees Must Be Paid Before Approval

Purpose:
Prevent Enrollment from being approved until fees are paid.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Validation Rules → New.
3. Rule Name: Fees_Must_Be_Paid_Before_Approval.
4. Error Condition Formula: AND(
   ISPICKVAL(Enrollment_Status__c, "Approved"),
   Fees_Paid__c = FALSE
)

5.Error Message: Fees must be paid before approving an enrollment.
6.Error Location: Top of Page (or field as needed).
Click Save.



Milestone 6 – Approval Process

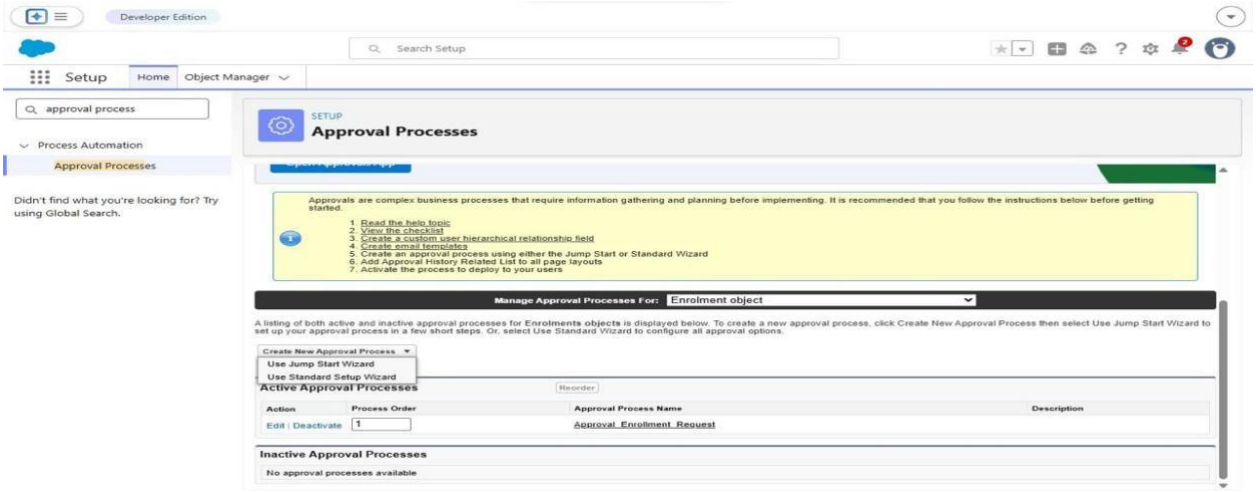
Activity 1: Create Approval Process for Enrollment Requests

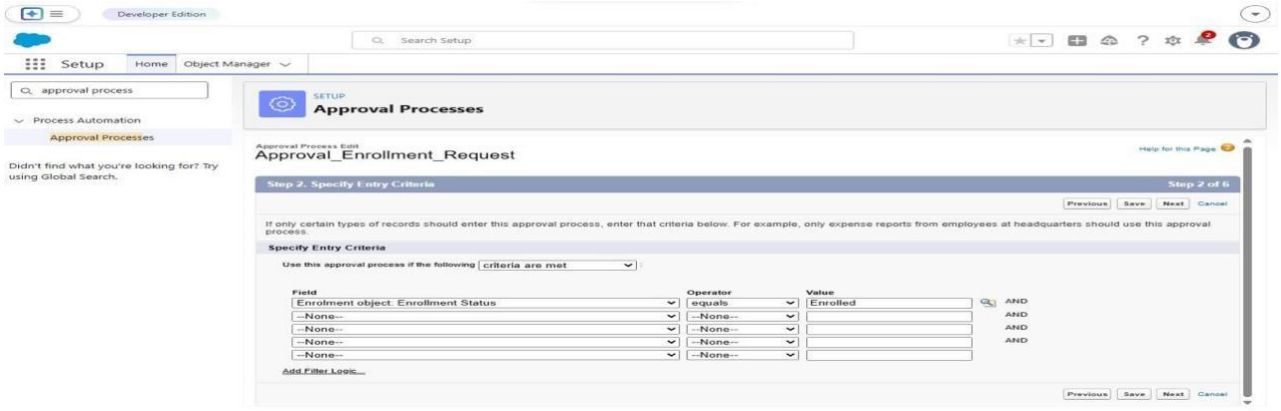
Purpose:
Route Enrollment requests to Training Manager for approval or rejection.

Steps:

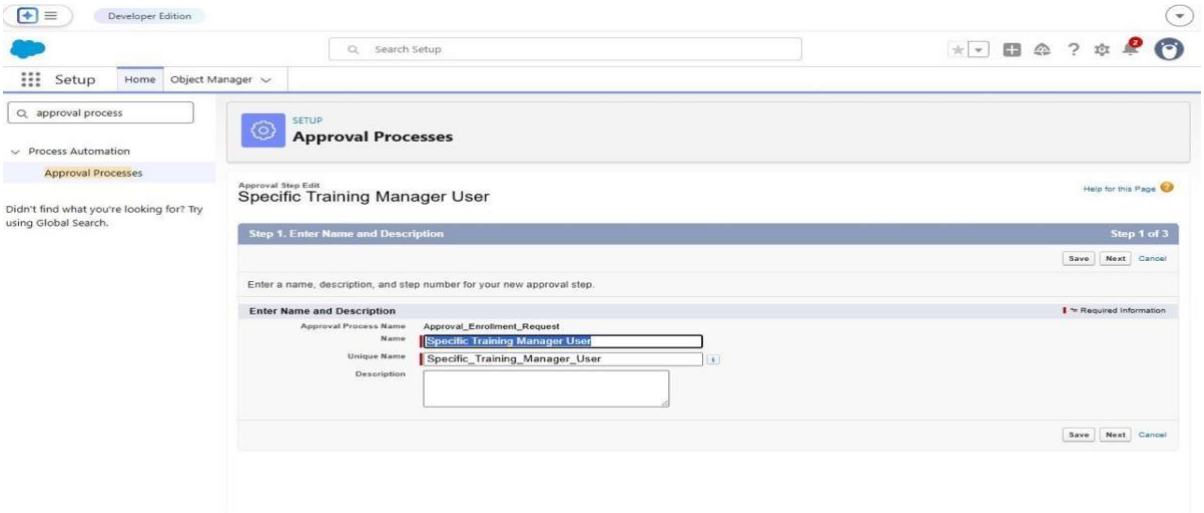
1. Go to Setup → Approval Processes → select Enrollment__c.
2. Click Create New Approval Process → Use Standard Setup.
3. Process Name: Approval_Enrollment_Request.

4. Entry Criteria: Enrollment_Status__c = "Requested".
5. Save and configure steps.

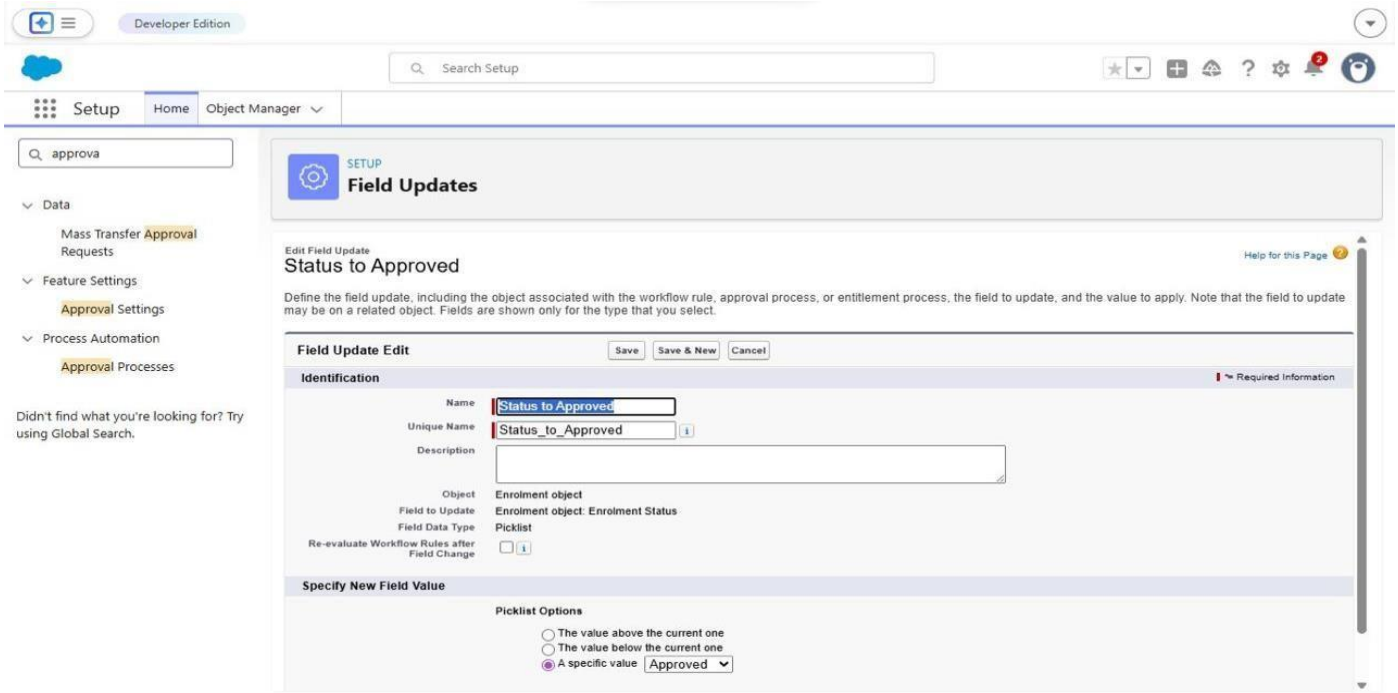




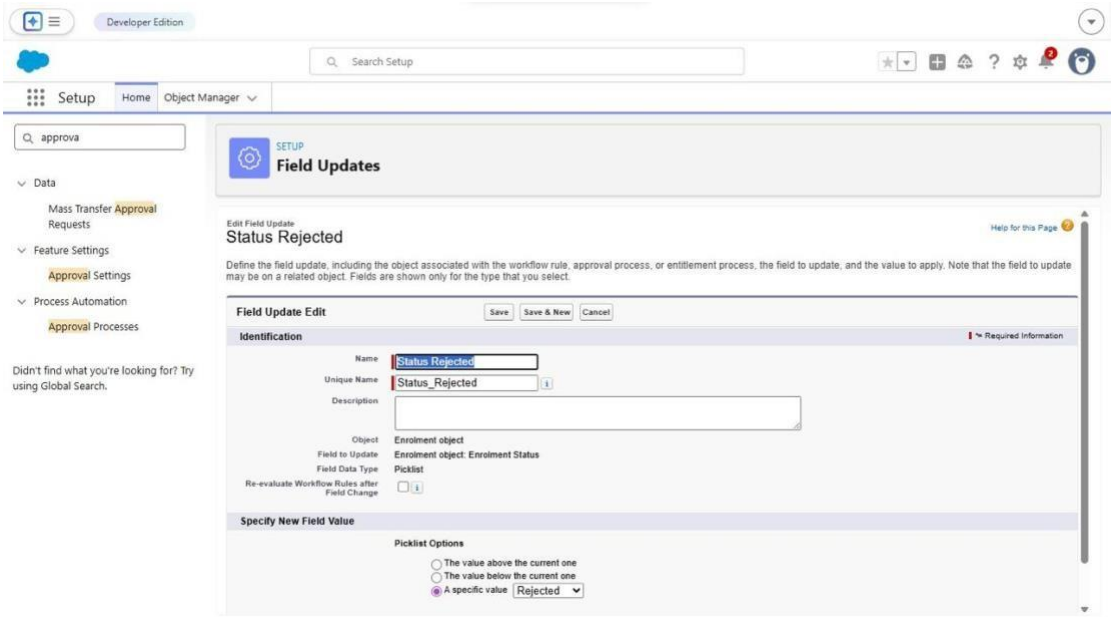
6. Add Approval Step:
7. Approver: specific Training Manager user.



8. Under Final Approval Actions:
9. Add Field Update to set Enrollment_Status__c = "Approved".



10. Under Final Rejection Actions:
11. Add Field Update to set Enrollment_Status__c = "Rejected".



12. Add Email Alert to send rejection email to student.

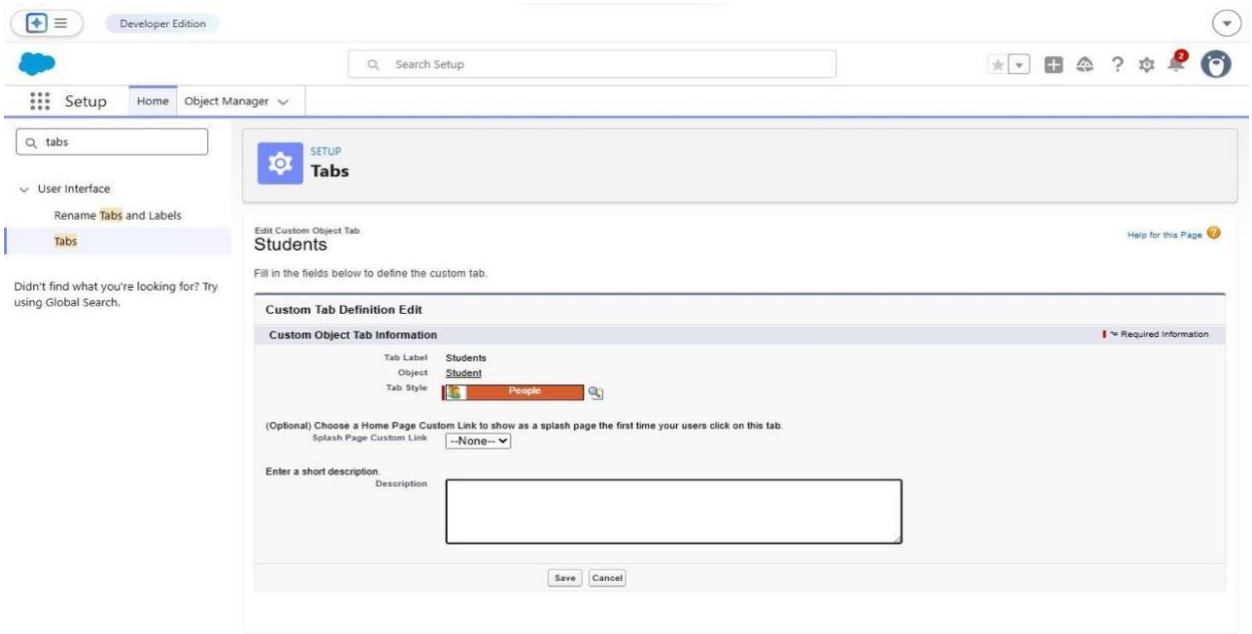
13. Activate the approval process.

Milestone 7- Tabs

Activity 1: Creating a Tab for the Student Object:

Go to Setup.

1. In the Quick Find box, type Tabs.
2. Under Custom Object Tabs, click New.
3. Select Student from the object list.
4. Choose a tab style (icon).
5. Click Next, assign it to all profiles, and then click Save.



Activity 2: Creating Remaining Tabs

Now create the Tabs for the remaining Objects, they are “Courses, Instructors, Enrollment”.
Follow the same steps as mentioned in Activity -1

Milestone 8 - The Lightning App

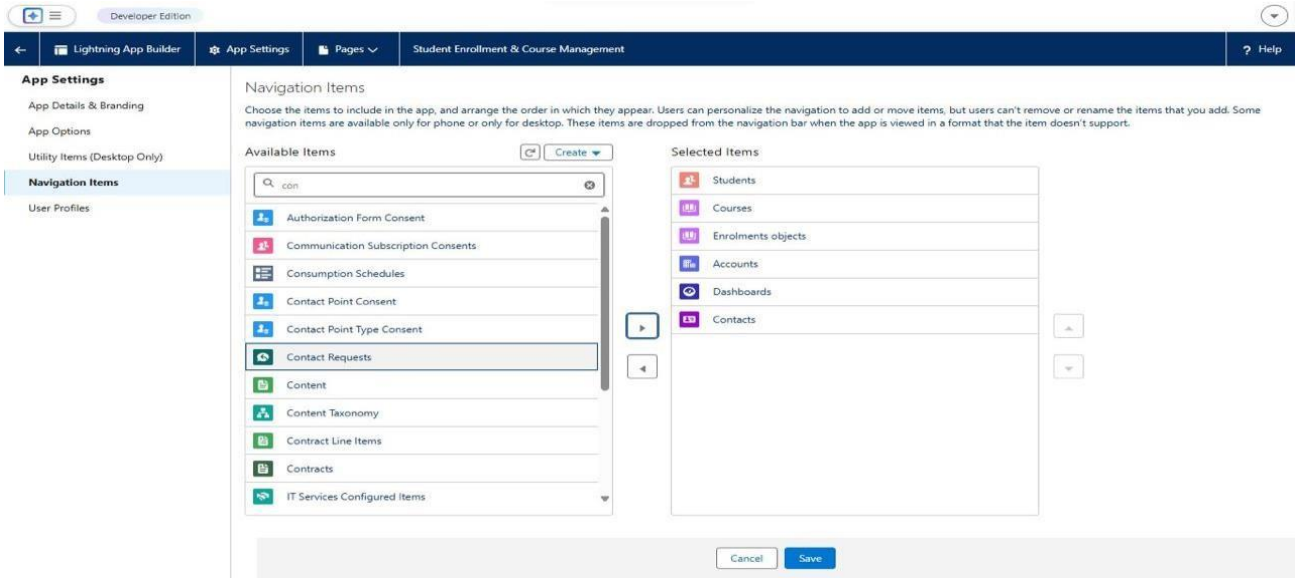
Activity 1: Creating the Lightning Application for the Project

Navigate to Setup.

Enter App Manager in the Quick Find search box.
Click New Lightning App.
Provide the app details:
App Name: Student Enrollment & Course Management
Developer Name auto-fills
Description: Optional, but can describe the purpose of the app
Click Next and choose the following settings:
App Branding (optional): Add logo or select color theme
Navigation Style: Choose Standard Navigation
Click Next and assign the app to specific profiles (e.g., **Admin, Enrollment Officer, Instructor**).
Click Next to select the items (tabs) to be included in the app. Add:

1. Student
2. Course
3. Instructor
4. Enrollment
5. Reports
6. Dashboards
7. Home
8. Tasks

Click Next and review the app settings.
Finally, click Save & Finish.



Milestone 9 – Page Layouts and Record:

Activity 1: Create Record Types for Enrollment:

Purpose:
To separate New Enrollment vs Re-Enrollment processes.
Steps:
1.Go to Setup → Object Manager → Enrollment__c.
2.Click Record Types → New.
3.For the first record type:

- Existing Record Type: Master
- Record Type Label: New Enrollment
- Enable for required profiles (Admin, Enrollment Officer). ● Select the default Page Layout → Save & New.

4.For the second record type:

- Record Type Label: Re-Enrollment ● Enable for same profiles. ● Select the default Page Layout → Save.

Activity 2: Separate Page Layo Createuts for Each Enrollment Record Type

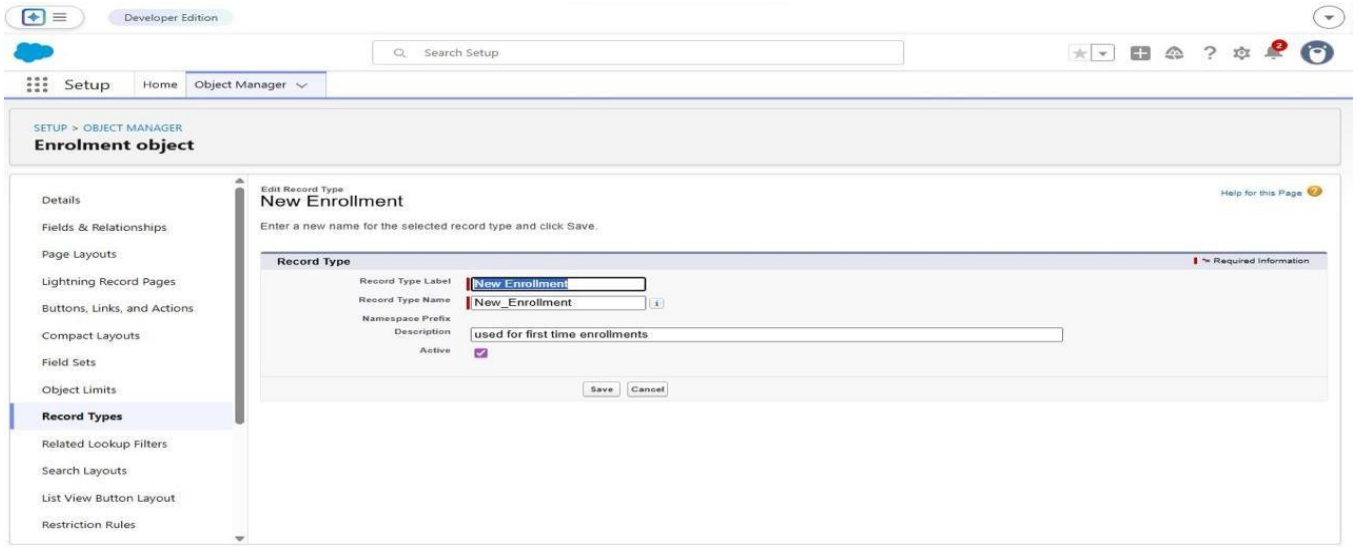
Purpose:

To show different fields for New Enrollment vs Re-Enrollment.

Steps:

1. Enrollment__c object → click Page Layouts.
2. Click New to create a layout for New Enrollment:
   - Clone from existing Enrollment Layout → name it New Enrollment Layout → Save.
   - Keep fields relevant for first-time enrollment: Student, Course, Instructor, Enrollment Date, Status, Fees, etc.
   - Remove unnecessary re-enrollment fields if any. ● Click Save.

3. Again, click New to create Re-Enrollment Layout:
   - Clone existing layout → name it Re-Enrollment Layout → Save.
   - Include fields like previous Enrollment reference (if any), ReEnrollment Reason, Status, Fees, etc. ● Click Save.



4. Click Page Layout Assignment → Edit Assignment.
5. For each profile and record type:

   - New Enrollment → use New Enrollment Layout.
6. Re-Enrollment → use Re-Enrollment Layout.

Click Save.

Activity 3: Enrollment R Addelated Lists to Parent Objects:

Purpose:
To see all Enrollments from Student, Course, and Instructor records. a) Student → Enrollment Related List Go to Setup → Object Manager → Student__c.
Click Page Layouts → open the main layout.
In the palette, select Related Lists.
Drag Enrollments related list onto the layout.
Customize columns if needed → Save. b)
Course → Enrollment Related List
Go to Object Manager → Course__c → Page Layouts.
Open layout → add Enrollments related list.
Save.
c) Instructor → Enrollment Related List
Go to Object Manager → Instructor__c → Page Layouts. Open layout → add Enrollments related list. Save.

Milestone 10 – Automation with Flow

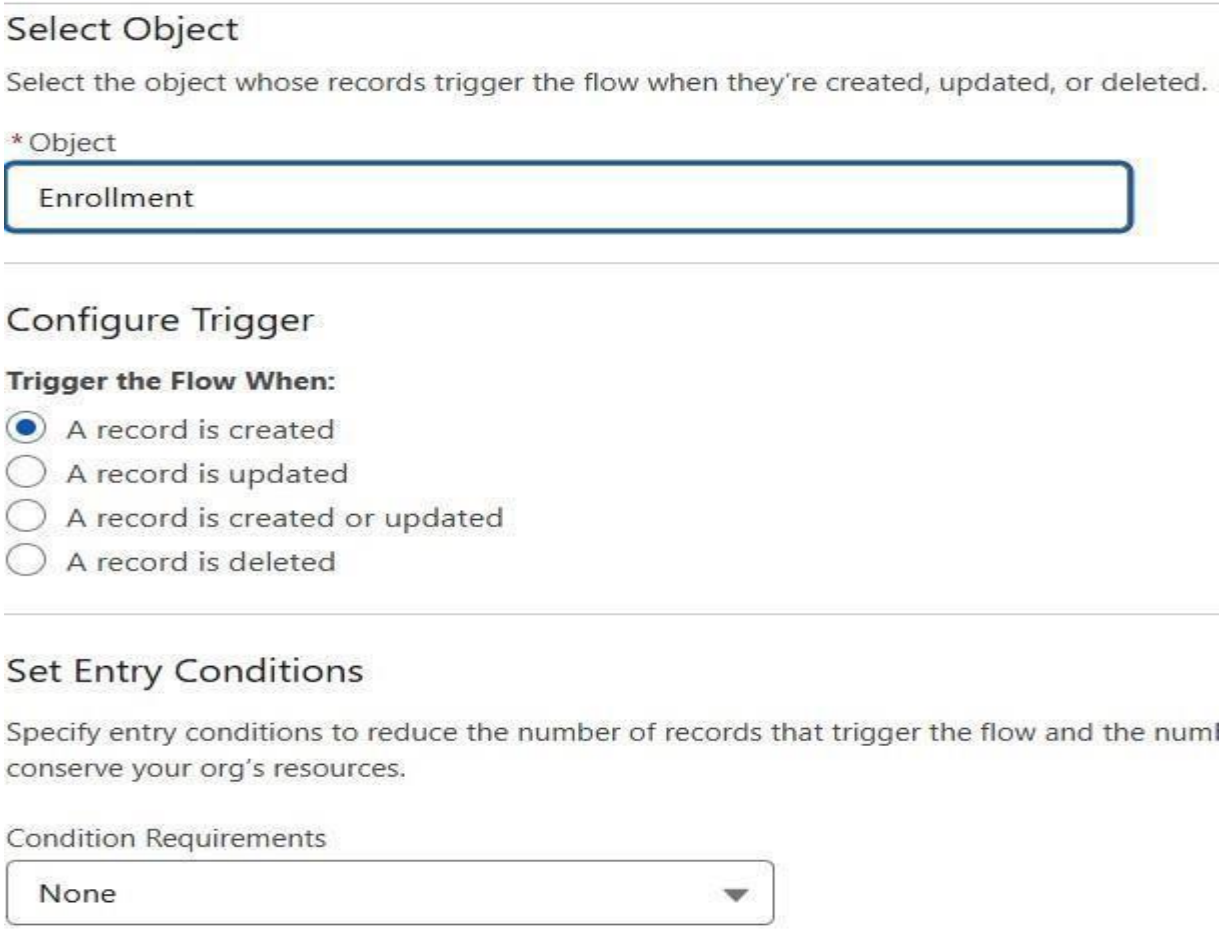
Activity 1: Auto-Populate Enrollment Date (Record-Triggered Flow)

Purpose:
Set Enrollment_Date__c to today when Enrollment is created.

Steps:

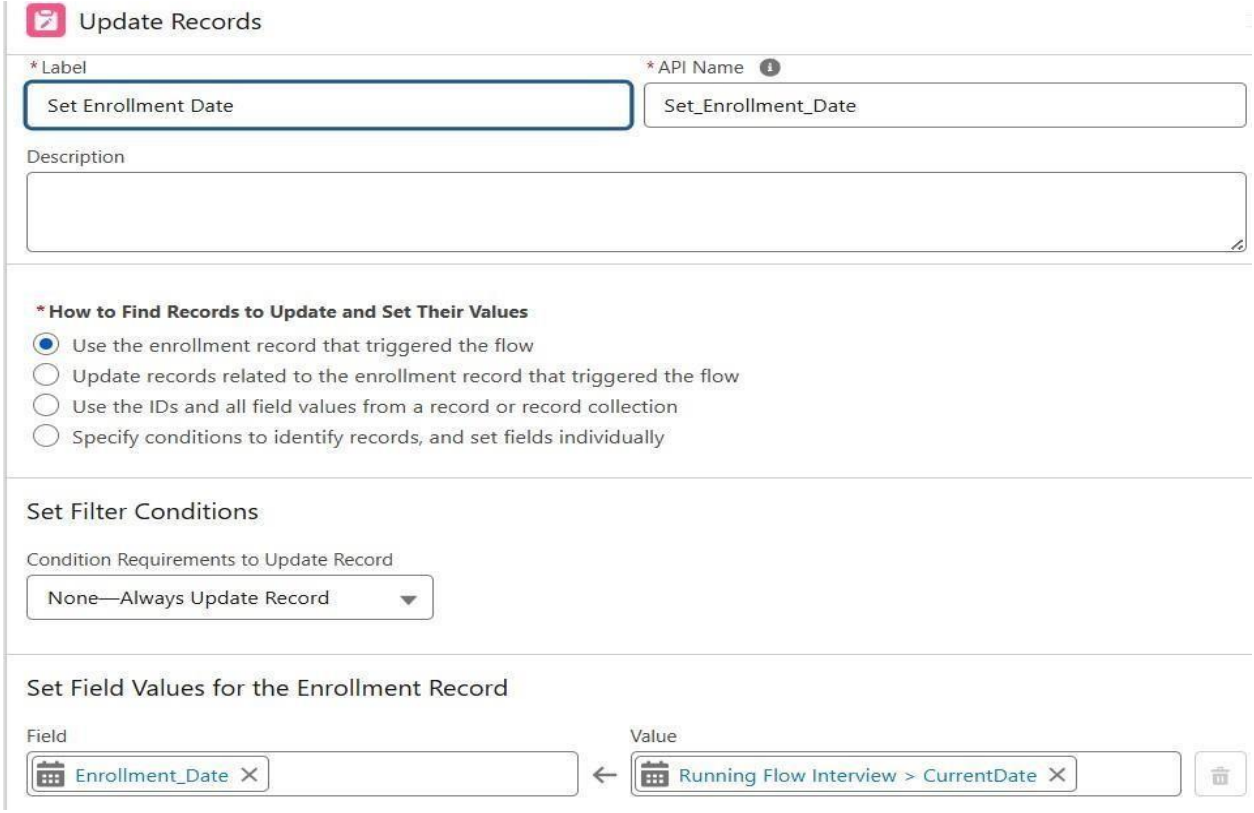
1. Go to Setup → Flows → New Flow.
2. Select Record-Triggered Flow → Create.
3. Object: Enrollment__c.
4. Trigger: A record is created.

5. Optimize for Actions and Related Records → Done.



6. Add Update Records element: ● Label: Set Enrollment Date.
   ● Use record that triggered the flow.
● Field: Enrollment_Date__c → Value: $Flow.CurrentDate.

7. Save as SetEnrollmentDate → Activate.



Activity 2: Send Email When Enrollment Approved

Purpose:
When Enrollment_Status__c = Approved, send email, alert admin, and set Student to Active.
Steps (high level for document):

1. Create Email Template and Email Alert for student notification.

2. Go to Flows → New → Record-Triggered Flow.
3. Object: Enrollment__c, Trigger on update.
4. Condition: Enrollment_Status__c Equals "Approved".

Configure Start

Enrollment

Configure Trigger

Trigger the Flow When:

- ○ A record is created
- ○ A record is updated
- ◉ A record is created or updated
- ○ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.

If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.

Condition Requirements

All Conditions Are Met (AND)

Field	Operator	Value
Enrollment Status	Equals	Approved

5. Add Get Records to fetch related Student__c.

Get Records

* Label: Get Student

* API Name: Get_Student

Description

Get Records of This Object

*** Data Source**

- ○ Data Cloud Object
- ◉ Salesforce Object

* Object: Student

Filter Student Records

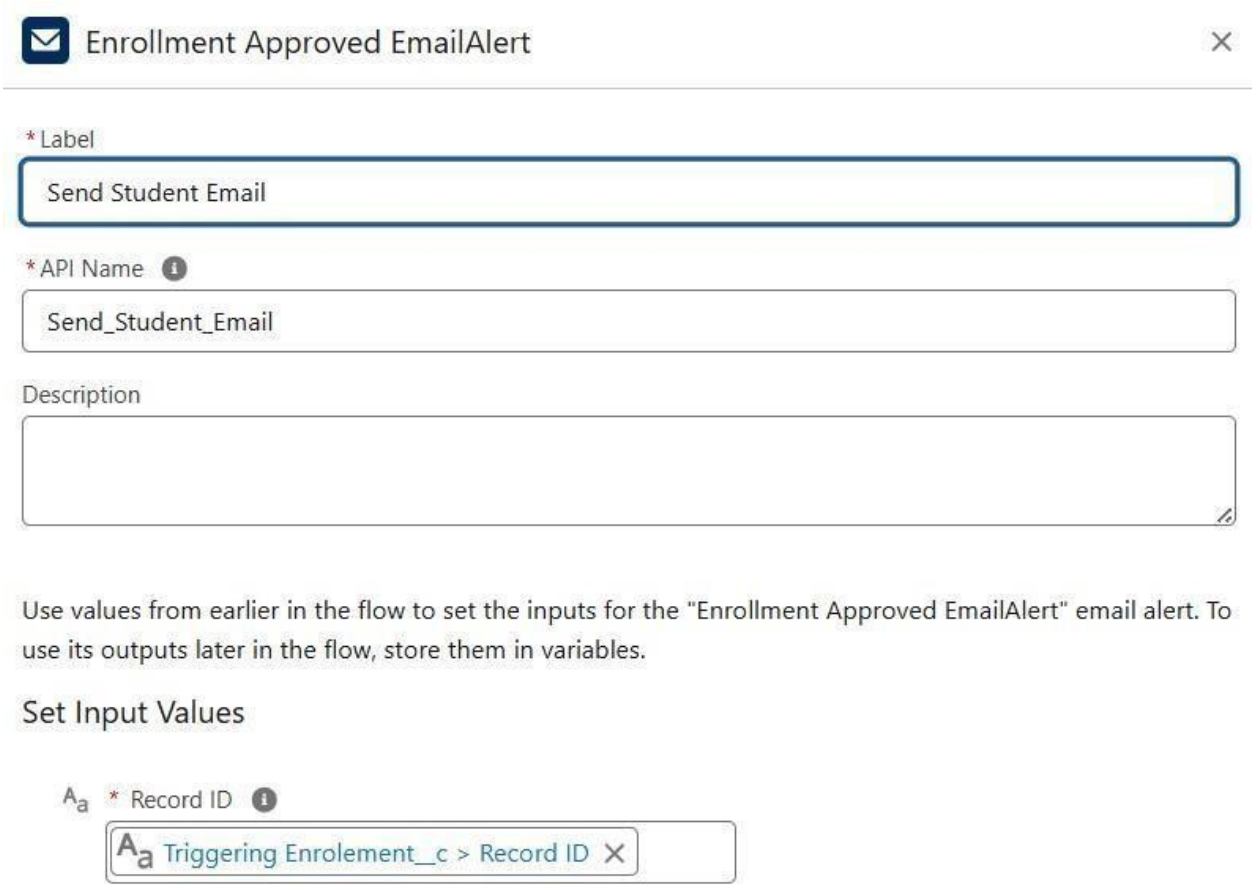
Condition Requirements

All Conditions Are Met (AND)

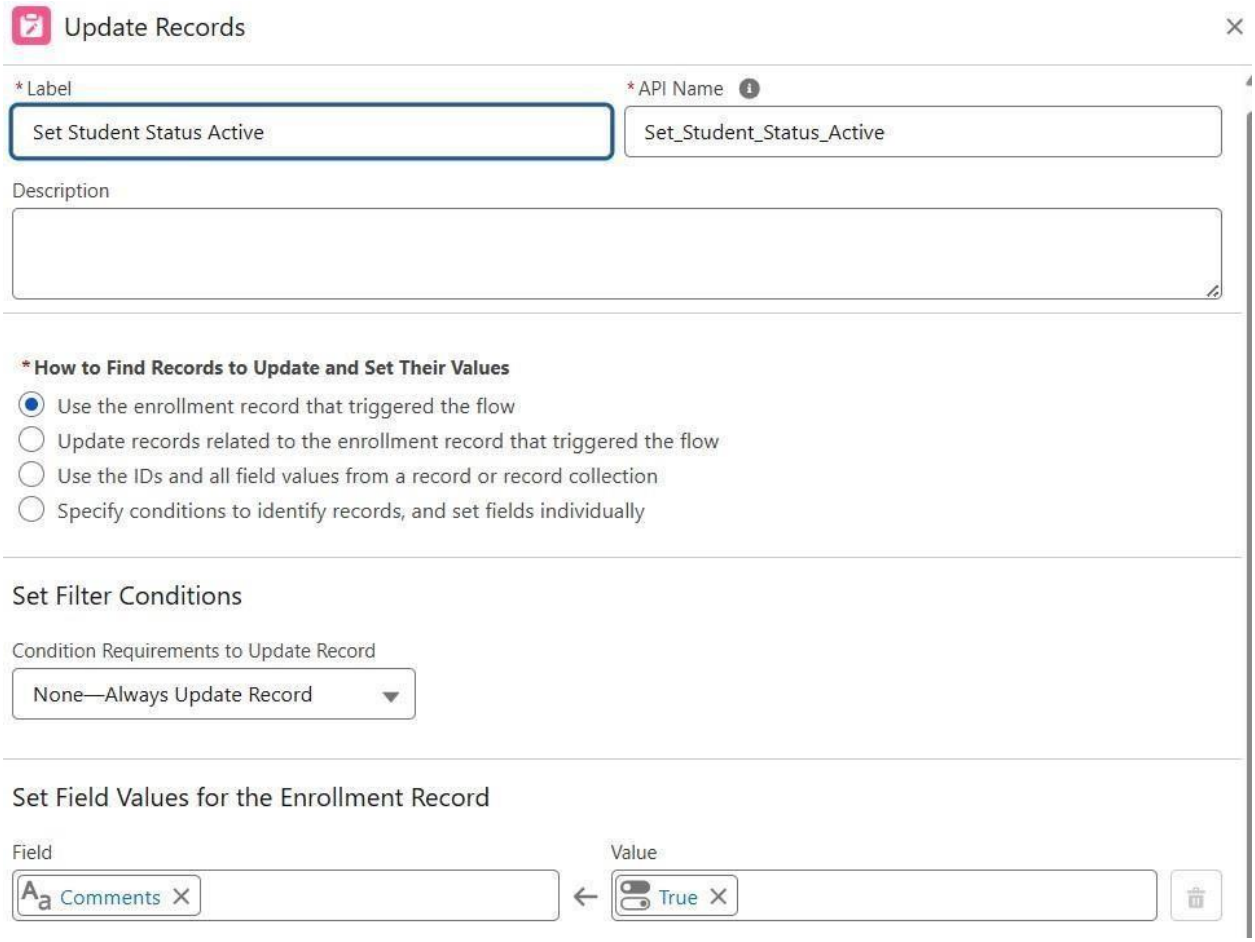
Field	Operator	Value
Status	Equals	Active

+ Add Condition

6. Add Action (Email Alert) to send email to student.



7. Add Action to notify Admin if needed.
8. Add Update Records to set Student__c.Status__c = "Active".



9. Save as Enrollment Approved Action → Activate.

Activity 3: Auto-Assign Instructor Based on Course Category

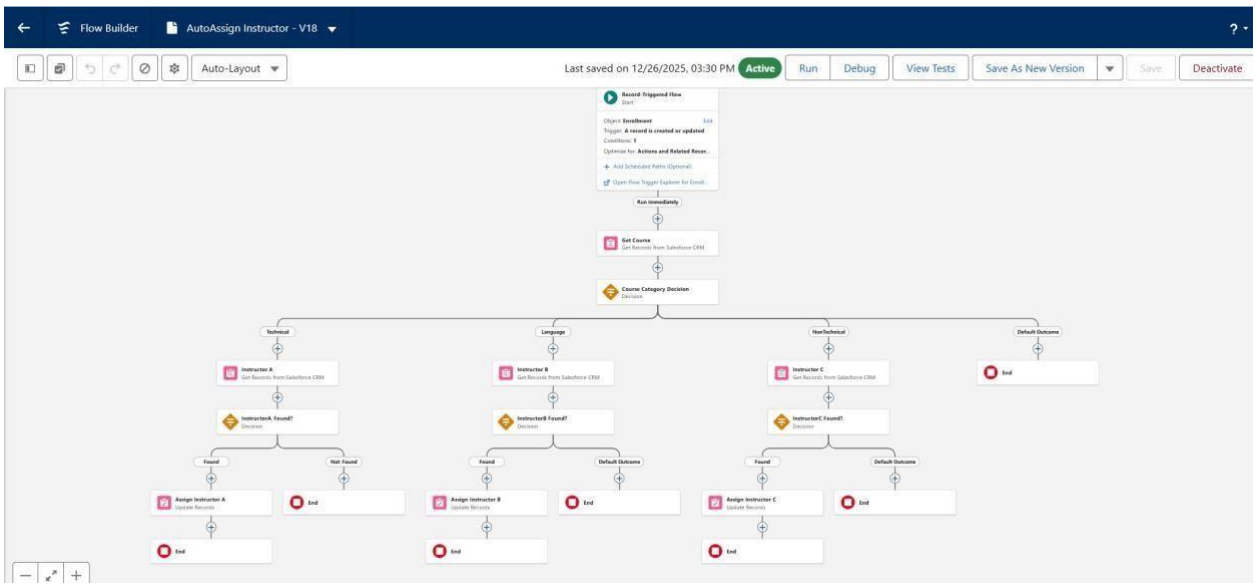
Purpose:
Assign Instructor automatically based on Course category.

Rules:

Technical → Instructor A
Language → Instructor B NonTechnical
→ Instructor C Steps:

1. Prepare Instructor__c records with codes (e.g., INSTR_A, INSTR_B, INSTR_C).

2. Create Record-Triggered Flow on Enrollment__c (created/updated).
3. Add Get Records to fetch related Course__c.
4. Add Decision element on Course__r.Category__c:
5. Technical / Language / Non-Technical outcomes.
6. For each outcome:
7. Get Instructor record (A/B/C) based on code or name.
8. Update Records: set Instructor__c = Get_Instructor_X.Id.
9. Save as AutoAssign_Instructor → Activate.

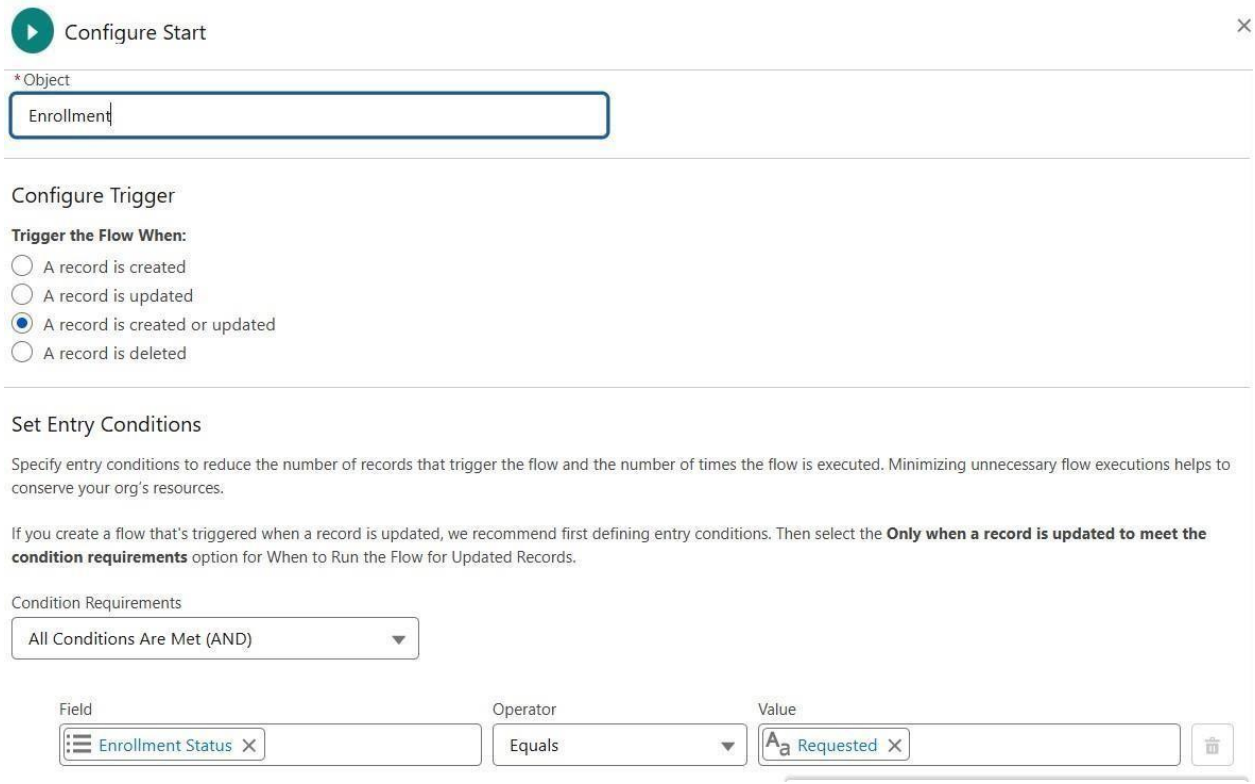


Activity 4: Auto-Create Task for Follow-Up

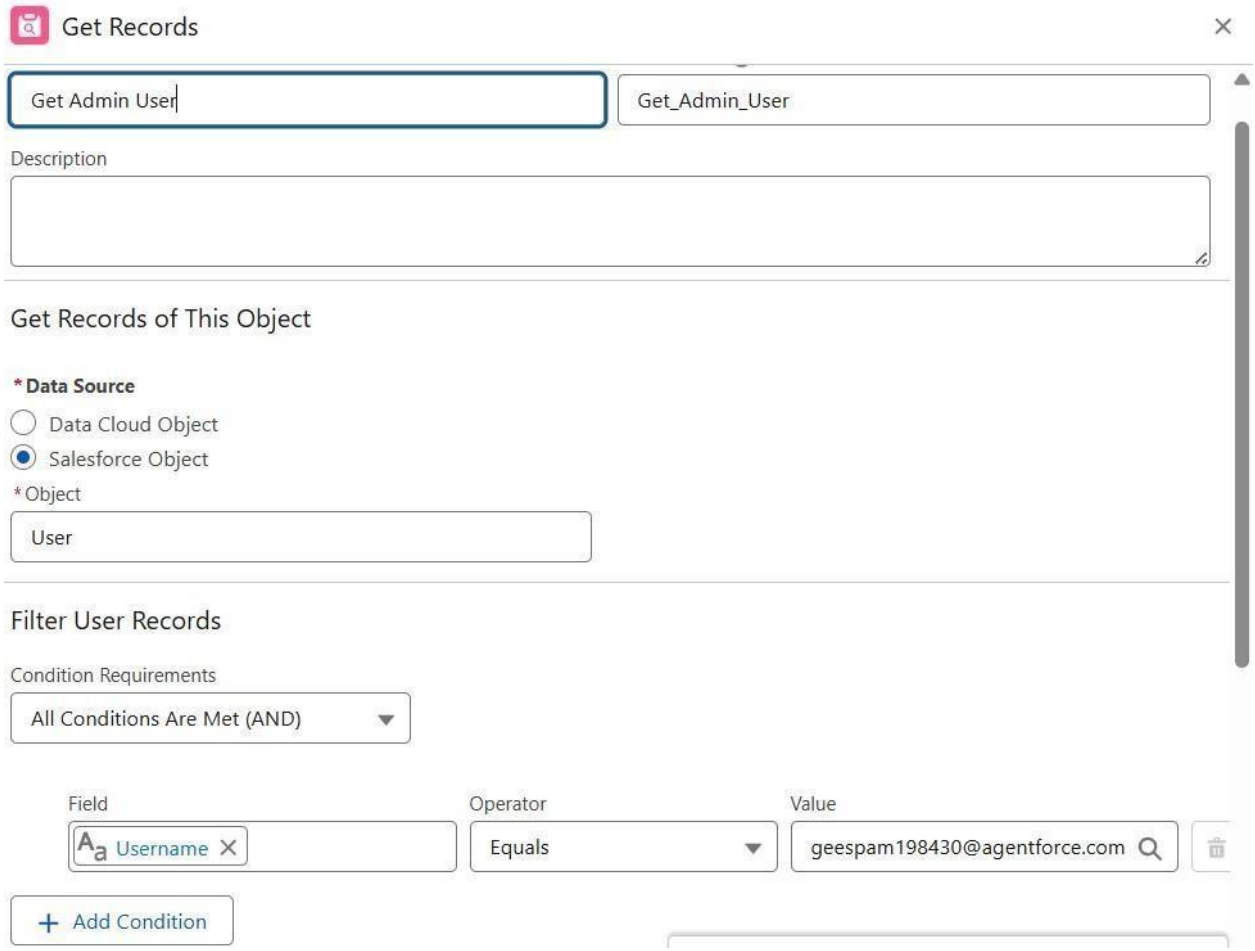
Purpose:
When Enrollment is “Requested”, create a follow-up task for Admin.

Steps:

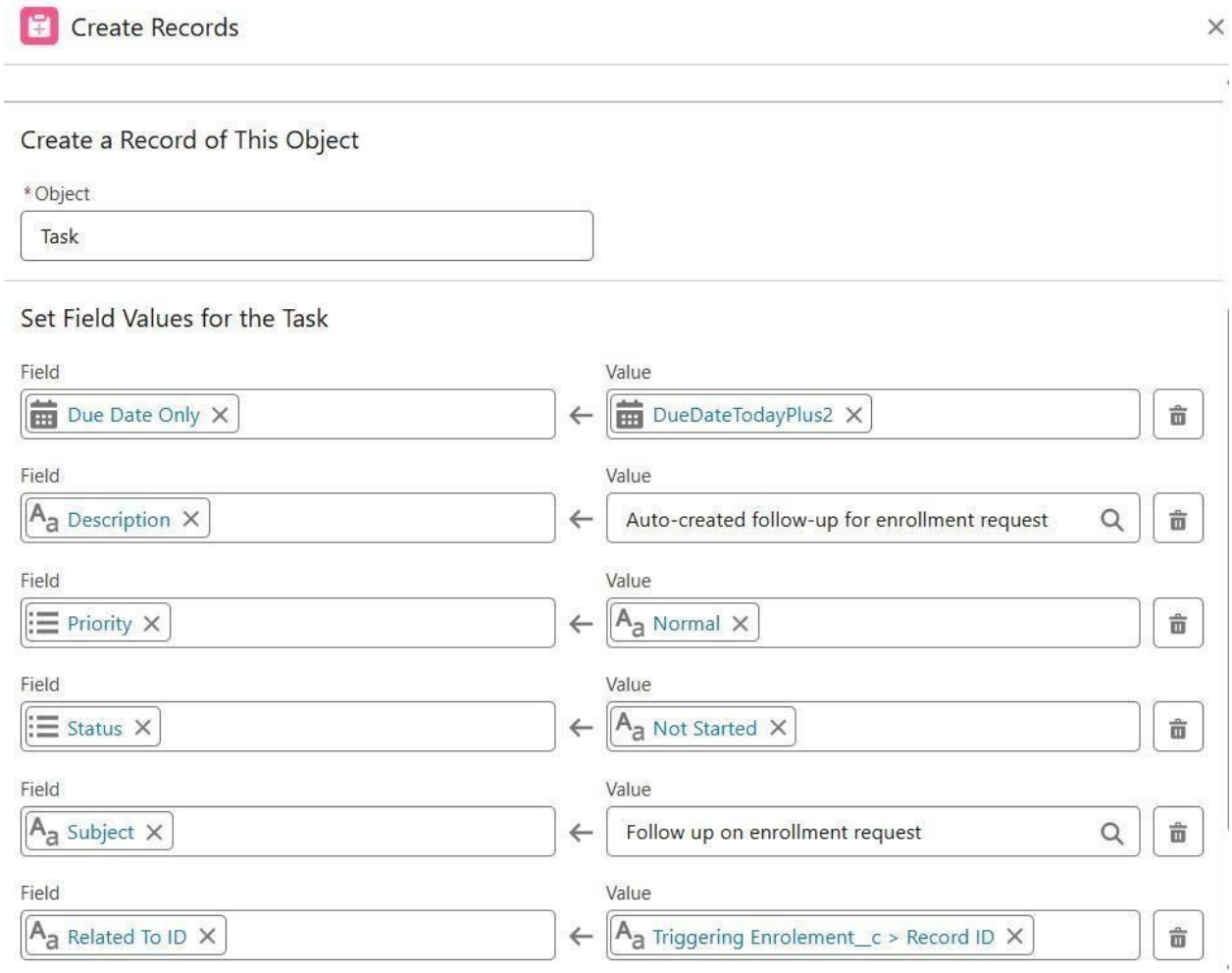
1. Create Record-Triggered Flow on Enrollment__c.
2. Trigger: Record created or updated; condition Enrollment_Status__c = "Requested".



3. Add Get Records to fetch Admin user (by Username / Profile).
4. Add Formula (Date): TODAY() + 2 for due date.



5. Add Create Records for Task:
6. Subject: Follow up on enrollment request.
7. OwnerId: Admin user Id.
8. WhatId: Enrollment Id.
9. ActivityDate: formula Today + 2.



10. Save as Create_Followup_Task → Activate.

Milestone 11 – Reports:

Activity 1: Create "Students by Status" Report

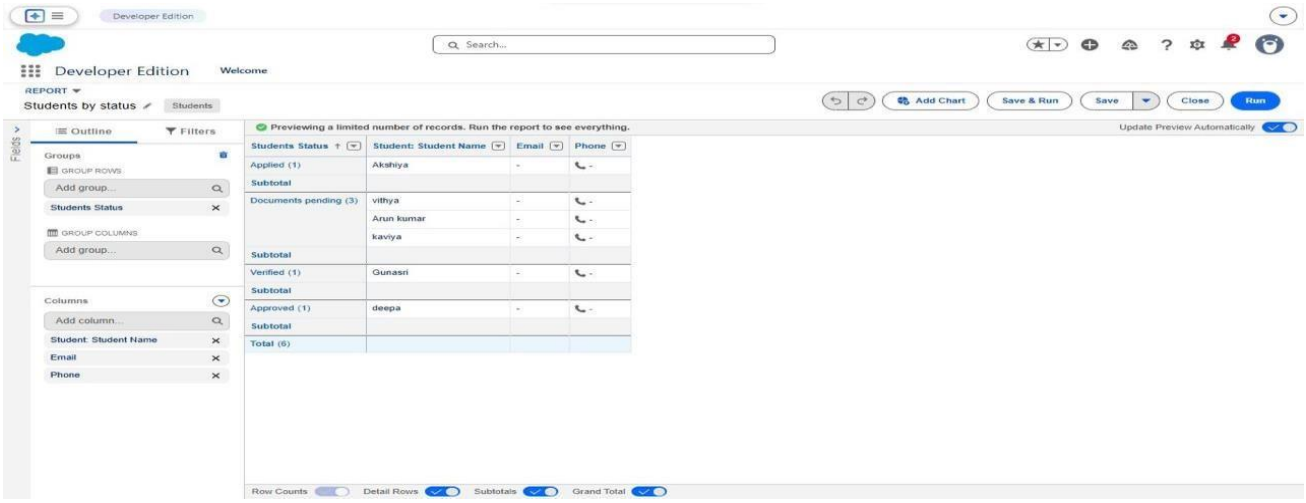
1.Go to **Reports** → **New Report**.

2.Report Type: Students.

3.Group rows by Status__c.

4.Add columns as needed (Name, Email, Phone).

5.Save as Students by Status.



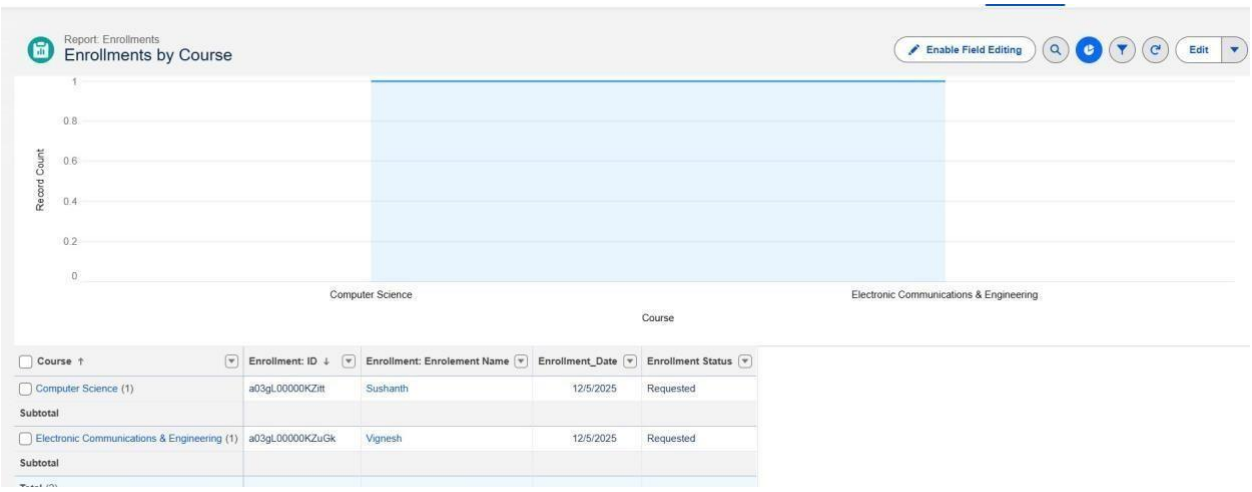
Activity 2: Create “Enrollments by Course” Report

1.New Report → Report Type: Enrollments.

2.Grouprows by Course__r.Name.

3.Show Record Count per course.

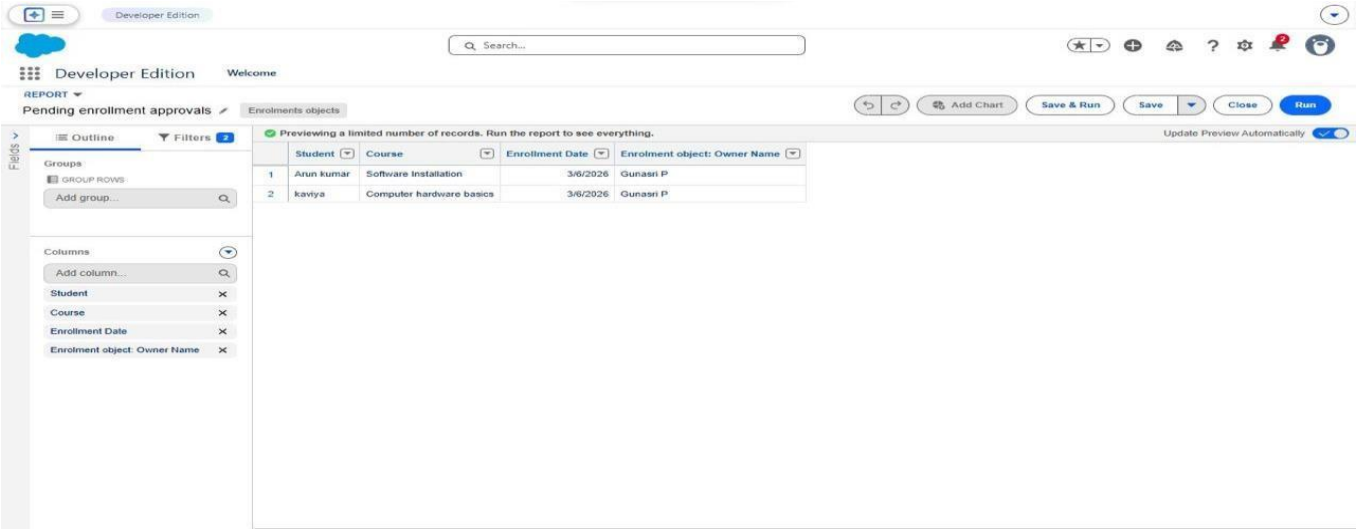
4.Save as Enrollments by Course



Activity 3: Create “Pending Enrollment Approvals” Report

1. New Report → Enrollments report type.
2. Filter: Enrollment_Status__c = "Requested".

3. Add columns (Student, Course, Owner, Created Date).
4. Save as Pending Enrollment Approvals.



Activity 4: Create "Revenue Report (Fees Paid)" Report

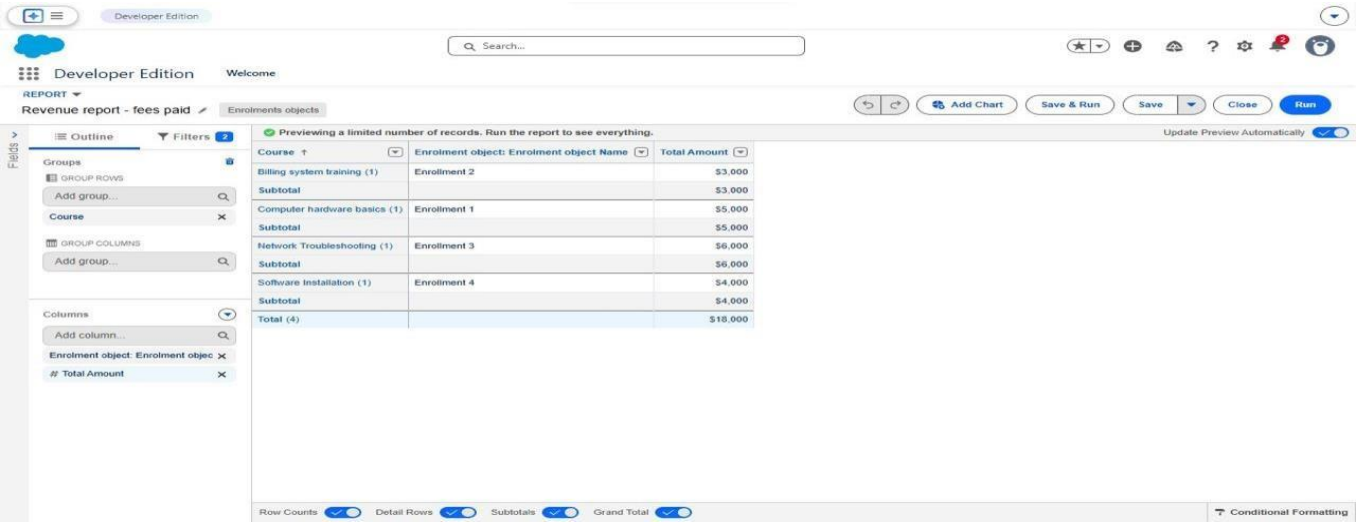
1.New Report → Enrollments report type.

2.Filter: Fees_Paid__c = TRUE.

3.Group by Course__r.Name.

4.Summarize Course_Fee__c or similar fee field.

5.Save as Revenue Report – Fees Paid.



Milestone 12 – Dashboards:

Activity 1: Create "Student Managemet Dnashboard"

Steps:

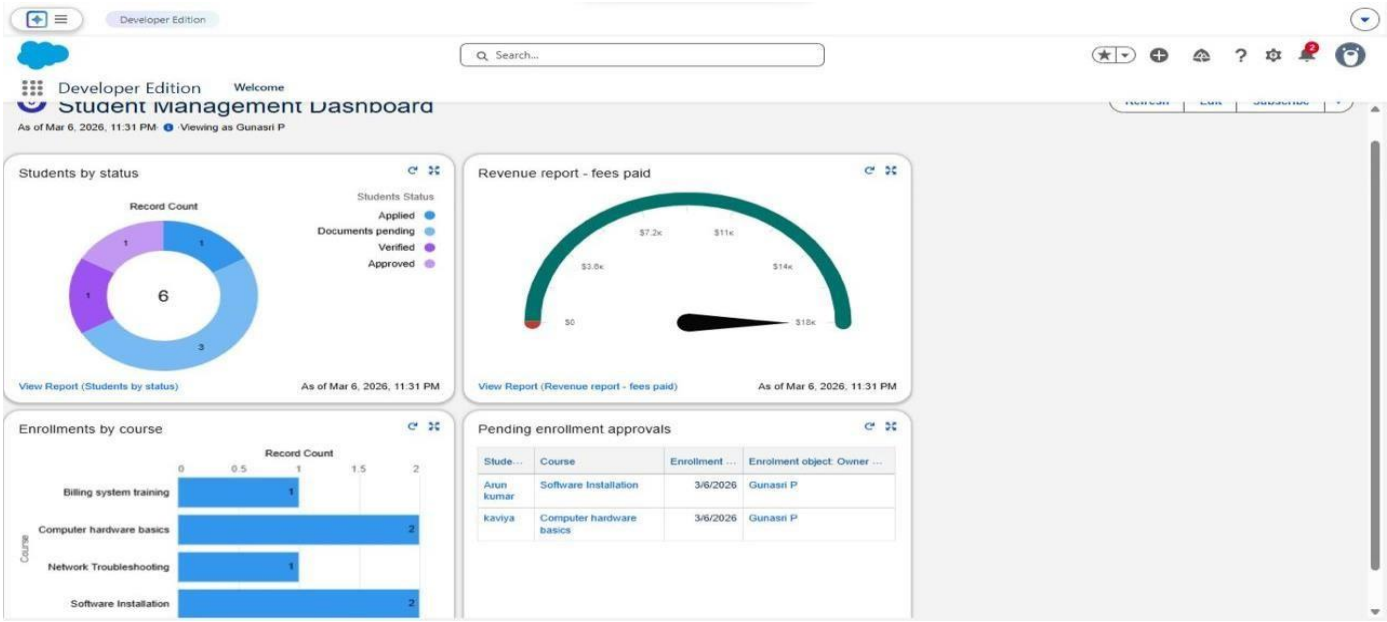
1. Go to **Dashboards → New Dashboard**.
2. Name: Student Management Dashboard.

3. Select folder → **Create**.
4. Select new Dashboard folder and Save.

Add components:

- **Donut Chart** → using Students by Status report.
- **Bar Chart** → using Enrollments by Course report.
- **Gauge** → using Revenue Report – Fees Paid (sum of revenue field).
- **Table** → using Pending Enrollment Approvals report.

4. Arrange components on canvas.



Click **Save → Done**.

Milestone 13 – Security Setup

Activity 1: Configure

Profiles):

1. Go to **Setup → Profiles → New** (or Clone**Profiles Required:**

- **Admin** – Full access.
- **Enrollment Officer** – Create/Update Enrollments.
- **Instructor** – Read-only access to assigned Enrollments.

2. **Steps (example for Enrollment Officer** fName: Enrollment Officer.
3. In **Object Settings**:
   - Enrollment__c: Read, Create, Edit.
   - Student__c / Course__c: Read (and Edit if required).
4. Save.

Repeat similarly for **Instructor** profile with **Read-Only** on Enrollments

Activity 2: Create Permission Sets

1. Go to **Setup → Permission Sets → New**.
2. Create Custom Object Access (additional access to project objects if needed).
3. Create Dashboard Editor (Create/Customize Reports & Dashboards permissions).
4. Assign to relevant users.

Activity 3: Configure Sharing & OWD

Steps:

1. Go to **Setup → Sharing Settings**.
2. Set **Organization-Wide Defaults**:
   - **Students (Student__c):** Public Read/Write.
   - **Courses (Course__c):** Public Read/Write.
   - **Enrollments (Enrollment__c):** Private.
3. Save.

For Instructor access to assigned Enrollments, optionally:

- Add a **User lookup** on Enrollment (Instructor User).
- Use a simple **Record-Triggered Flow** to set **OwnerId = Instructor User** so they can see only their own records.

7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce user interface to verify the proper functioning of all implemented features of the Student Enrollment & Course Management System (SECMS). These include student, course, instructor, and enrollment record creation; validation rule checks for mandatory fields, eligibility, and enrollment constraints; verification of record type behavior for different course categories; flow execution for automated enrollment approvals and seat availability checks; and Apex trigger functionality to ensure automatic confirmation and record updates. Additionally, screenshots of generated reports and dashboards were documented to validate accurate data visualization, enrollment tracking, and performance monitoring. Submit the Screenshots while testing.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Dependence on the salesforce platform
- Initia set
- Continous internet access is required

9. Conclusion:

The Student Enrollment and Course Management System developed using Salesforce provides an efficient solution for managing academic enrollment operations. The system integrates student data, course management, instructor assignment, and reporting within a single platform.

10. FUTURE SCOPE

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org